DATA SCIENCE CAPSTONE PROJECT 3 IDEAS

By Irina Kim

Idea 1: The K-Beauty Spa Architect (MVP)

- **Core Tech:** NLP (Semantic Search) + Recommendation Systems (Tiered) + Vision (Skin Analysis).
- **The Problem:** Friends and clients find K-Beauty at Costco/Sephora confusing.
- **The Solution:** A one-stop assistant that provides a daily routine (Cleanser -> Serum -> Moisturizer -> SPF) and optionally schedules Botox/Microneedling based on the skin's "Longevity Cycle."

Idea 2: The Menopause Transition Monitor (The "Clinical Play")

- **Core Tech:** Time Series Analysis + NLP.
- **The Problem:** Symptoms like brain fog and sleep issues are intermittent and hard to track.
- **The Solution:** Analyze longitudinal data (SWAN study) to classify which stage of menopause a user is in.
- *Image of the stages of menopause and hormonal shifts over time*
- **NLP Use:** Sentiment analysis of user logs to detect "Hormonal Dips" before they manifest as severe physical symptoms.

Idea 3: The Oura Vitality Engine (The "N-of-1" Play)

- **Core Tech:** Biometric Time Series + Recommendation Systems.
- **The Problem:** Oura/Apple Watch data tells you what happened yesterday, but not what to do today.
- **The Solution:** Merges daily biometric time-series (HRV, Sleep, RHR) with the NIH Supplement Database.
- **The System:** Detects a 3-day recovery dip and recommends a specific K-Beauty "Self-Care Night" (e.g., Ginseng mask + Magnesium) to restore baseline readiness.